



Apex Institute of Technology

Computer Science & Engineering

Experiment 9

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Semester: 4

Date of Performance: 17.04.2026

**Subject Name: Database
Management System**

Subject Code: 24CSH-298

AIM: To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

OBJECTIVES:

- To design and implement a PL/SQL package that includes procedures and shared cursors for structured and modular program development.

SOFTWARE REQUIREMENTS:

- Database Management System:
 - Oracle
- Database Administration Tool:
 - Oracle SQL Developer

PRACTICAL/EXPERIMENT STEPS:

1. An employees table was created to store employee details such as ID, name, salary, and department.
2. Sample records were inserted into the employees table for testing purposes.
3. A PL/SQL package specification was created to declare procedures related to employee data.
4. A package body was developed to implement the logic of the declared procedures.
5. A shared cursor was defined inside the package to retrieve employee records efficiently.
6. A procedure display_all_employees was implemented to display all employee details using the cursor.



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7. A procedure `display_employee` was created to fetch details of a specific employee using employee ID.
8. The package procedures were executed to verify correct functionality and output.

PROCEDURE:

1. Oracle SQL environment (SQL Developer / FreeSQL) was opened and the required database was selected.
2. The employees table was created using the `CREATE TABLE` command.
3. Sample data was inserted into the table using `INSERT` statements.
4. A package specification was created using `CREATE OR REPLACE PACKAGE` to declare procedures.
5. A package body was created using `CREATE OR REPLACE PACKAGE BODY` to define procedures and a shared cursor.
6. Procedures were written to display all employees and to retrieve a specific employee by ID.
7. The package procedures were executed using `BEGIN ... END;` blocks.
8. The output was displayed using `DBMS_OUTPUT.PUT_LINE` and verified for correctness.

CODE:

```
CREATE TABLE employees (  
    emp_id NUMBER PRIMARY KEY,  
    emp_name VARCHAR2(50),  
    salary NUMBER,  
    department VARCHAR2(50)  
);  
  
INSERT INTO employees VALUES (101, 'Rahul', 50000, 'HR');  
INSERT INTO employees VALUES (102, 'Anita', 60000, 'Finance');  
INSERT INTO employees VALUES (103, 'Karan', 55000, 'IT');  
INSERT INTO employees VALUES (104, 'Neha', 65000, 'Marketing');  
  
COMMIT;  
  
CREATE OR REPLACE PACKAGE emp_package AS  
    PROCEDURE display_all_employees;  
    PROCEDURE display_employee(p_emp_id NUMBER);  
  
END emp_package;  
  
/  
  
CREATE OR REPLACE PACKAGE BODY emp_package AS  
    CURSOR emp_cursor IS  
        SELECT emp_id, emp_name, salary, department
```



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```
FROM employees;

PROCEDURE display_all_employees IS
BEGIN
    FOR emp_rec IN emp_cursor LOOP
        DBMS_OUTPUT.PUT_LINE(
            emp_rec.emp_id || ' | ' ||
            emp_rec.emp_name || ' | ' ||
            emp_rec.salary || ' | ' ||
            emp_rec.department
        );
    END LOOP;
END display_all_employees;

PROCEDURE display_employee(p_emp_id NUMBER) IS
    v_emp employees%ROWTYPE;
BEGIN
    SELECT * INTO v_emp
    FROM employees
    WHERE emp_id = p_emp_id;

    DBMS_OUTPUT.PUT_LINE(
        v_emp.emp_id || ' | ' ||
        v_emp.emp_name || ' | ' ||
        v_emp.salary || ' | ' ||
        v_emp.department
    );

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found.');
```

END display_employee;

```
END emp_package;
/

SET SERVEROUTPUT ON;

BEGIN
    emp_package.display_all_employees;
END;
/

BEGIN
    emp_package.display_employee(102);
END;
/
```



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I/O ANALYSIS:

1. Create a table

The employees table is successfully created with fields such as emp_id, emp_name, salary, and department. The emp_id is defined as a primary key, ensuring unique employee records.

```
SQL> CREATE TABLE employees (  
    emp_id NUMBER PRIMARY KEY,  
    emp_name VARCHAR2(50),  
    salary NUMBER,...  
Show more...
```

Table EMPLOYEES created.

Elapsed: 00:00:00.020

2. Insert Records

Sample employee records are inserted into the table successfully. The table now contains multiple employee entries for testing the package procedures.

```
SQL> INSERT INTO employees VALUES (104, 'Neha', 65000, 'Marketing') SQL> COMMIT
```

1 row inserted.

Elapsed: 00:00:00.002

Commit complete.

Elapsed: 00:00:00.001

3. Creating Package Specification

The package specification emp_package is created successfully. It declares procedures to display all employees and to fetch a specific employee using ID.

```
SQL> CREATE OR REPLACE PACKAGE emp_package AS  
    -- Procedure to display all employees  
    PROCEDURE display_all_employees;  
    -- Procedure to display employee by ID...  
Show more...
```

Package EMP_PACKAGE compiled

Elapsed: 00:00:00.011



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4. Creating Package Body with Shared Cursor

The package body is created successfully. A shared cursor (emp_cursor) is defined to retrieve employee records efficiently and is used across procedures.

```
SQL> CREATE OR REPLACE PACKAGE BODY emp_package AS
        CURSOR emp_cursor IS
            SELECT emp_id, emp_name, salary, department
            FROM employees;...
Show more...
```

Package Body EMP_PACKAGE compiled

Elapsed: 00:00:00.027

5. Display Employee Details (All & Specific)

The procedures display_all_employees and display_employee are executed successfully. All employee records are displayed using the shared cursor, and specific employee details are retrieved correctly using employee ID.

101	Rahul	50000	HR	
102	Anita	60000	Finance	
103	Karan	55000	IT	
104	Neha	65000	Marketing	

102	Anita	60000	Finance
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PL/SQL procedure successfully completed. PL/SQL procedure successfully completed.

Elapsed: 00:00:00.129

Elapsed: 00:00:00.008

LEARNING OUTCOMES:

1. Understood the concept and structure of PL/SQL packages.
2. Differentiated between package specification and package body.
3. Developed packages containing procedures and shared cursors.
4. Wrote modular and reusable PL/SQL code.
5. Applied package development techniques used in organizations such as Rippling, Oracle, SAP, and PayPal.